After grinding the educts CuO, Y_3O_3 and $BaCO_3$ to a fine powder, their diffraction spectrum is measured with a X-Ray diffraction spectrometer. The resulting spectrum is seen in Figure 1.

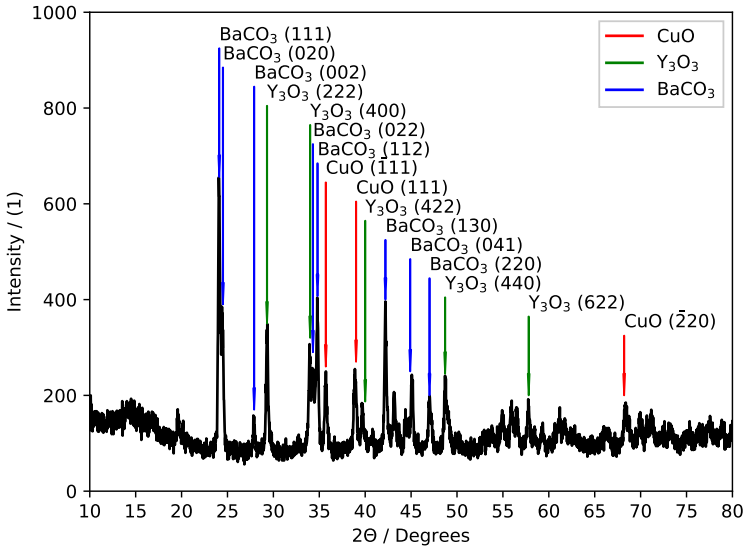


Figure 1: Spectrum before sintering with labeld peaks for material and plane.

A whole array of peaks can be identified. They are tabled in Tables 1 to 3 and can be seen on the figure. Especially for higher angles, a whole sea of tightly packed peaks are measured. As their angle only varies by up to a degree, they can not be labled with certainty, as it is very easy to confuse them with each other. Nonethereless two prominent peaks of CuO and Y_3O_3 are identified.

$2\Theta/^\circ$	(hkl)
29.1	(222)
33.8	(400)
39.8	(422)
48.5	(440)
57.6	(622)

Table 1: Standard diffraction peaks for cubic Y_2O_3

$2\Theta/^\circ$	(hkl)
35.5	$(\bar{1}11)$
38.8	(111)
68.0	$(\bar{2}20)$

Table 2: Standard diffraction peaks for CuO

$2\Theta/^\circ$	(hkl)
23.9	(111)
24.3	(020)
27.7	(002)
34.6	(112)
42.0	(130)
44.7	(041)
46.8	(220)

Table 3: Standard diffraction peaks for orthorhombic $BaCO_3$

After sintering them at 950 C for 12 h, the ceramic is grinded and with diffraction measured again. As a result the spectrum of Figure 2 is obtained.

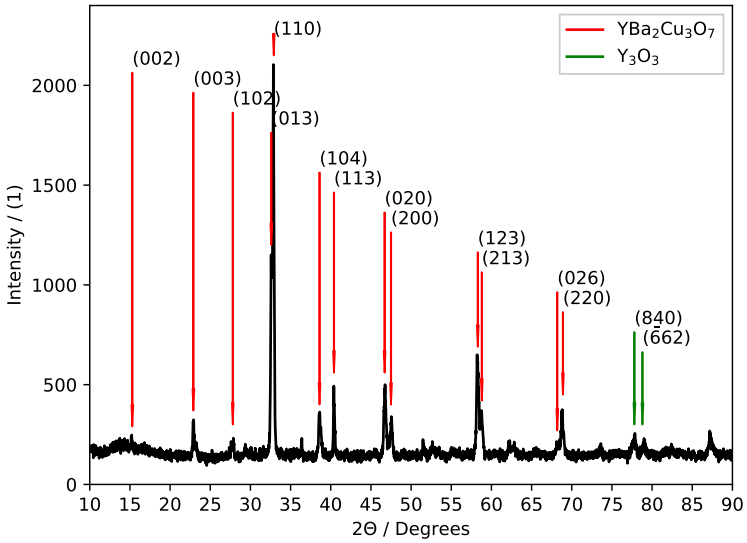


Figure 2: Spectrum after sintering with labeld peaks for material and plane.

It can be seen that most peaks are identified with planes of the $YBa_2Cu_3O_7$ ceramic. They are tabled in Table 4. Some speaks for higher angles can not be assigned to the new ceramic. Instead they lie directly on two high angles diffractions peaks of the Y_2O_3 educt. This shows, that there are still some residuals that did not react yet. This can happen because larger crumbs in

the mixture, i.e. the grinding process was not thorough enough, the sintering time was to short or the temperature was to low. The assigned angles are listed in Tables 4 to 5. Unfortunately, the high angle peaks of Y_2O_3 can not be seen with certain in Figure 1, because the dense sea of diffraction peaks does not allow for differentiation, as already mentioned.

$2\Theta/^\circ$	(hkl)
15.3	(002)
22.9	(003)
27.8	(102)
32.6	(013)
32.9	(110)
38.6	(104)
40.4	(113)
46.7	(020)
47.5	(200)
58.3	(123)
58.8	(213)
68.2	(026)
68.9	(220)

Table 4: Standard diffraction peaks for orthorhombic $YBa_2Cu_3O_7$

$2\Theta/^\circ$	(hkl)
77.8	(840)
78.8	$(\bar{6}62)$

Table 5: High-angle Y_2O_3 secondary-phase peaks

Afterwards, using a hydraulic press, the powder is pressed into pallets and sintered again. The final product is the high-temperature superconductor $YBa_2Cu_3O_7$.